AP PRECALCULUS · TOPIC 1.7

Rational Functions and End Behavior

Section 1 — Multiple Choice
Circle the letter of the correct answer **SOLVED**

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $f(x) = \dfrac{3x^2+1}{x^2-5}$. Which line is the horizontal asymptote of the graph of f?

A. $y = 0$ **B.** $y = 3$

C. $y = -\dfrac{1}{5}$ **D.** $x = \sqrt{5}$

B. Equal degrees: the ratio of leading coefficients is $\dfrac{3}{1} = 3$.

2 Let $g(x) = \dfrac{2x+1}{x^2+4}$. What is $\lim\limits_{x\to\infty} g(x)$?

A. 2 **B.** $\dfrac{1}{4}$

C. ∞ **D.** 0

D. The denominator has the larger degree, so the quotient approaches 0.

3 Let $h(x) = \dfrac{x^3-1}{2x^2+5}$. Which statement is true?

A. $h(x) \to \infty$ as $x \to \infty$ and $h(x) \to -\infty$ as $x \to -\infty$

B. $h(x) \to \infty$ as $x \to \infty$ and as $x \to -\infty$

C. $h(x) \to \dfrac{1}{2}$ as $x \to \infty$ and as $x \to -\infty$

D. $h(x) \to 0$ as $x \to \infty$ and as $x \to -\infty$

A. Leading terms give $\dfrac{x^3}{2x^2} = \dfrac{x}{2}$, which grows without bound with the sign of x.

4 Let $r(x) = \dfrac{4-x^2}{2x^2+x}$. What is $\lim\limits_{x\to-\infty} r(x)$?

A. $\dfrac{1}{2}$ **B.** 2

C. $-\dfrac{1}{2}$ **D.** 4

C. Leading terms $\dfrac{-x^2}{2x^2} = -\dfrac{1}{2}$, written in any order of terms.

5 Which function has a horizontal asymptote at $y = 2$?

A. $f(x) = \dfrac{2x+1}{x^2+1}$ **B.** $f(x) = \dfrac{x^2+2}{x-3}$

C. $f(x) = \dfrac{x+6}{3x^2}$ **D.** $f(x) = \dfrac{6x^2-x}{3x^2+5}$

D. Equal degrees with ratio $\dfrac{6}{3} = 2$. The others give 0, no horizontal asymptote, and 0.

6 A rational function $r(x) = \dfrac{p(x)}{q(x)}$ satisfies $\lim\limits_{x\to\infty} r(x) = 0$. Which must be true?

A. The degree of p equals the degree of q

B. The degree of p is less than the degree of q

C. The degree of p is greater than the degree of q

D. $p(x)$ has no real zeros

B. A limit of 0 occurs only when the denominator has the larger degree.

7 Let $r(x) = \dfrac{x}{x^2+1}$. Which statement is true?

A. The line $y = 0$ is a horizontal asymptote, and the graph never touches it

B. The line $y = 1$ is a horizontal asymptote

C. The line $y = 0$ is a horizontal asymptote, and the graph crosses it at $(0, 0)$

D. The graph has no horizontal asymptote

C. The denominator's degree is larger, so $y = 0$ is the asymptote; since $r(0) = 0$ the graph meets it. A horizontal asymptote describes large $|x|$ only.

8 Let $f(x) = \dfrac{2x+3}{x-1}$. As $x \to \infty$, how does $f(x)$ approach its horizontal asymptote $y = 2$?

A. From above

B. From below

C. It oscillates around $y = 2$

D. It never approaches $y = 2$

A. $f(x) = 2 + \dfrac{5}{x-1}$, and $\dfrac{5}{x-1} > 0$ for large positive x.

Rational Functions and End Behavior

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

After chlorine is added to a city water reservoir, its concentration in mg/L is modeled by $C(t) = \dfrac{4t+3}{t^2+2}$, where $t \geq 0$ is the number of hours since the chlorine was added.

(a) Find $\lim_{t\to\infty} C(t)$. Interpret the value in context. DOK3 2 pt

The denominator has the larger degree, so $\lim_{t\to\infty} C(t) = 0$: in the long run the chlorine concentration approaches 0 mg/L.

(b) Find the greatest concentration and the time it occurs, to three decimal places. DOK3 2 pt

Maximum $C \approx 2.351$ mg/L at $t \approx 0.851$ hours.

(c) After the peak, find the first time the concentration drops to 0.5 mg/L. Give your answer to three decimal places. DOK3 2 pt

Solve $\dfrac{4t+3}{t^2+2} = 0.5$: $t^2 - 8t - 4 = 0$, so $t = 4 + \sqrt{20} \approx 8.472$ hours.

Question 10 No calculator · 6 points

Let $f(x) = \dfrac{5x^2 - x}{2x^2 + 3}$ and $g(x) = \dfrac{x^3 + 1}{x^2 - 4}$.

(a) Write $\lim_{x\to\infty} f(x)$ and give the equation of the horizontal asymptote of f. DOK3 2 pt

Equal degrees: $\lim_{x\to\infty} f(x) = \dfrac{5}{2}$, so the horizontal asymptote is $y = \dfrac{5}{2}$.

(b) Describe the end behavior of g at both ends, using limit notation. DOK3 2 pt

Leading terms give $\dfrac{x^3}{x^2} = x$: $\lim_{x\to\infty} g(x) = \infty$ and $\lim_{x\to-\infty} g(x) = -\infty$.

(c) Does g have a horizontal asymptote? Justify using degrees. DOK3 2 pt

No. The numerator degree (3) is greater than the denominator degree (2), so the outputs grow without bound.

Answer key (Section 1): 1B 2D 3A 4C 5D 6B 7C 8A